

LAB_014_Activity

April 1, 2025

1 Activity: Debug Python Code

1.1 Introduction

One of the biggest challenges faced by analysts is ensuring that automated processes run smoothly. Debugging is an important practice that security analysts incorporate in their work to identify errors in code and resolve them so that the code achieves the desired outcome.

Through a series of tasks in this lab, you'll develop and apply your debugging skills in Python.

Tips for completing this lab

As you navigate this lab, keep the following tips in mind:

- `### YOUR CODE HERE ###` indicates where you should write code. Be sure to replace this with your own code before running the code cell.
- Feel free to open the hints for additional guidance as you work on each task.
- To enter your answer to a question, double-click the markdown cell to edit. Be sure to replace the “[Double-click to enter your responses here.]” with your own answer.
- You can save your work manually by clicking File and then Save in the menu bar at the top of the notebook.
- You can download your work locally by clicking File and then Download and then specifying your preferred file format in the menu bar at the top of the notebook.

1.2 Scenario

In your work as a security analyst, you need to apply debugging strategies to ensure your code works properly.

Throughout this lab, you'll work with code that is similar to what you've written before, but now it has some errors that need to be fixed. You'll need to read code cells, run them, identify the errors, and adjust the code to resolve the errors.

1.3 Task 1

The following code cell contains a syntax error. In this task, you'll run the code, identify why the error is occurring, and modify the code to resolve it. (To ensure that it has been resolved, run the code again to check if it now functions properly.)

```
[5]: # For loop that iterates over a range of numbers
# and displays a message each iteration

for i in range(10)
    print("Connection cannot be established")
```

```
File "<ipython-input-5-324a3968fb71>", line 4
for i in range(10)
~
```

SyntaxError: invalid syntax

Hint 1

The header of a for loop in Python requires specific punctuation at the end.

Hint 2

The header of a for loop in Python requires a colon (:) at the end.

Question 1 What happens when you run the code before modifying it? How can you fix this?

[When running the code as is, the output shows a SyntaxError on line 4, where it says “for i in range(10). To fix this SyntaxError, you would need to add a colon after the parenthesis, so that it shows”for i in range(10):”. This way, the code will be fixed.]

1.4 Task 2

In the following code cell, you’re provided a list of usernames. There is an issue with the syntax. In this task, you’ll run the cell, observe what happens, and modify the code to fix the issue.

```
[8]: # Assign `usernames_list` to a list of usernames

usernames_list = ["djames", "jpark", "tbailey", "zdutchma "esmith", "srobinso",
↪ "dcoleman", "fbautist"]

# Display `usernames_list`

print(usernames_list)
```

```
File "<ipython-input-8-8a568c7729fd>", line 3
usernames_list = ["djames", "jpark", "tbailey", "zdutchma "esmith",
↪ "srobinso", "dcoleman", "fbautist"]
~
```

SyntaxError: invalid syntax

Hint 1

Each element in `usernames_list` is a username and should be a string. In Python, a string should have quotation marks around it.

Hint 2

When creating a list in Python, the elements of the list should be separated with commas. There should be a comma between every two consecutive elements.

Question 2 What happens when you run the code before modifying it? How can you fix it?

[When running the code as is, the output shows a `SyntaxError` on line 3. To fix this `SyntaxError`, the username “zdutchma”, should have a closing double quotes, along with a comma afterwards to fix the code.]

1.5 Task 3

In the following code cell, there is a syntax error. Your task is to run the cell, identify what is causing the error, and fix it.

```
[11]: # Display a message in upper case  
  
print("update needed".upper())
```

```
File "<ipython-input-11-b640f5ee0427>", line 3  
print("update needed".upper()  
      ^
```

```
SyntaxError: unexpected EOF while parsing
```

Hint 1

Calling a function in Python requires both opening and closing parentheses.

Hint 2

In the code above, check that each function call has both opening and closing parentheses.

Question 3 What happens when you run the code before modifying it? What is causing the syntax error? How can you fix it?

[When running the code as is, the output shows a `SyntaxError` on line 3, showing that there is an “unexpected EOF while parsing”. The syntax error is caused because it is missing another closing

parenthesis in the print() function, and to fix this, adding another “)” at the end of the line for the print() code will fix the syntax error.]

1.6 Task 4

In the following code cell, you’re provided a `usernames_list`, a `username`, and code that determines whether the username is approved. There are two syntax errors and one exception. Your task is to find them and fix the code. A helpful debugging strategy is to focus on one error at a time and run the code after fixing each one.

```
[17]: # Assign `usernames_list` to a list of usernames that represent approved users

usernames_list = ["djames", "jpark", "tbailey", "zdutchma", "esmith",
    ↪ "srobinso", "dcoleman", "fbautist"]

# Assign `username` to a specific username

username = "esmith"

# For loop that iterates over the elements of `usernames_list` and determines
    ↪ whether each element corresponds to an approved user

for name in username_list:

    # Check if `name` matches `username`
    # If it does match, then display a message accordingly

    if name = username:
        print("The user is an approved user")
```

The user is an approved user

Hint 1

In Python, the `=` assignment operator allows you to assign or reassign a variable to a value, and the `==` comparison operator allows you to compare one value to another (or the value of one variable to the value of another).

Hint 2

Indentation is important in Python syntax. Check that the indentation inside the `for` loop and the indentation inside the `if` statement are correct.

Hint 3

Check that each time a variable is used, it’s spelled in the same way it was spelled when it was assigned.

Question 4 What happens when you run the code before modifying it? What is causing the errors? How can you fix it?

[When running the code as is, the first error that is outputted is a `SyntaxError` on line 16. It shows that there is an invalid syntax at the equal sign “=”. To fix this, adding another “=” in between “name” and “username” should fix that one syntax error. The next syntax error that is outputted is an `IndentationError`, displaying that it was expecting an indented block inside the conditional “if” statement on line 17. Fixing this issue just requires indenting the `print()` function. Now, another issue that is outputted is a `NameError`, an exception where the “username_list” is not defined on line 11. To fix this exception, I looked at line 3 and noticed that the variable is spelled “usernames_list”, with an extra “s” after “username”. Deleting the extra “s” and running the code, will then fix all of the errors encountered, and to check if the logic is correct, which the code and output align.]

1.7 Task 5

In this task, you’ll examine the following code and identify the type of error that occurs. Then, you’ll adjust the code to fix the error.

```
[20]: # Assign `usernames_list` to a list of usernames

usernames_list = ["elarson", "bmoreno", "tshah", "sgilmore", "eraab"]

# Assign `username` to a specific username

username = "eraab"

# Determine whether `username` is the final username in `usernames_list`
# If it is, then display a message accordingly

if username == usernames_list[5]:
    print("This username is the final one in the list.")
```

This username is the final one in the list.

Hint 1

Recall that indexing in Python starts at 0.

Hint 2

Identify how many elements there are in the `usernames_list`.

Hint 3

Since indexing in Python starts at 0 and the `usernames_list` contains 5 elements, identify which index value corresponds to the final element in `usernames_list`.

Question 5 What happens when you run the code before modifying it? What type of error is this? How can you fix it?

[When running the code as is, the first error that is outputted is an `IndexError` exception error on line 12. This index error shows that the list index is out of range. To fix this error, changing the

index after “usernames_list” from “5” to “4”, should correct the index and output that message that is intended.]

1.8 Task 6

In this task, you’ll examine the following code. The code imports a text file into Python, reads its contents, and stores the contents as a list in a variable named `ip_addresses`. It then removes elements from `ip_addresses` if they are in `remove_list`. There are two errors in the code: first a syntax error and then an exception related to a string method. Your goal is to find these errors and fix them.

```
[23]: # Assign `import_file` to the name of the text file

import_file = "allow_list.txt"

# Assign `remove_list` to a list of IP addressess that are no longer allowed to
↪access the network

remove_list = ["192.168.97.225", "192.168.158.170", "192.168.201.40", "192.168.
↪58.57"]

# With statement that reads in the text file and stores its contents as a list
↪in `ip_addresses`

with open(import_file, "r") as file:
    ip_addresses = file.read()

# Convert `ip_addresses` from a string to a list

ip_addresses = ip_addresses.split()

# For loop that iterates over the elements in `remove_list`,
# checks if each element is in `ip_addresses`,
# and removes each element that corresponds to an IP address that is no longer
↪allowed

for element in remove_list:
    if element in ip_addresses:
        ip_addresses.remove(element)

# Display `ip_addresses` after the removal process

print(ip_addresses)
```

```
['ip_address', '192.168.25.60', '192.168.205.12', '192.168.6.9',
'192.168.52.90', '192.168.90.124', '192.168.186.176', '192.168.133.188',
'192.168.203.198', '192.168.218.219', '192.168.52.37', '192.168.156.224',
```

```
'192.168.60.153', '192.168.69.116']
```

Hint 1

A `with` statement in Python requires a colon (`:`) at the end of the header.

Hint 2

The `.split()` method in Python is used on strings to convert them to lists. To call the `.split()` method, place the string you want to split in front of the method call.

Question 6 What happens when you run the code before modifying it? What is causing the errors? How can you fix them?

[When running the code as is, the first error that is outputted is a `SyntaxError` involving invalid syntax on line 11. To fix this issue, inserting a colon at the end of the line will complete the “with” statement. The next error that is outputted after running the updated code, is a `NameError` exception, where it shows the name “split” is not defined on line 16. This is because the `.split()` option is not utilized correctly, and to fix this, the code should be rewritten to “`ip_addresses.split()`”. After running the code, we ensure that the logic is intended with the code, which was to remove elements that were in the “remove_list” from the “ip_addresses” list that was parsed. Checking the output with the “remove_list” shows that the code is running correctly.]

1.9 Task 7

In this final task, there are three operating systems: OS 1, OS 2, and OS 3. Each operating system needs a security patch by a specific date. The patch date for OS 1 is “March 1st”, the patch date for OS 2 is “April 1st”, and the patch date for OS 3 is “May 1st”.

The following code stores one of these operating systems in a variable named `system`. Then, it uses conditionals to output the patch date for this operating system.

However, this code has logic errors. Your goal is to assign the `system` variable to different values, run the code to examine the output, identify the error, and fix it.

```
[29]: # Assign `system` to a specific operating system as a string

system = "OS 2"

# Assign `patch_schedule` to a list of patch dates in order of operating system

patch_schedule = ["March 1st", "April 1st", "May 1st"]

# Conditional statement that checks which operating system is stored in
# `system` and displays a message showing the corresponding patch date

if system == "OS 1":
    print("Patch date:", patch_schedule[2])

elif system == "OS 2":
```

```
print("Patch date:", patch_schedule[0])

elif system == "OS 3":
    print("Patch date:", patch_schedule[2])
```

Patch date: March 1st

Hint 1

Recall that indexing in Python starts at 0.

Hint 2

Note that the patch dates in `patch_schedule` are in order of operating system. The first patch date in `patch_schedule` corresponds to OS 1, the second patch date in `patch_schedule` corresponds to OS 2, and so on.

Hint 3

Since indexing in Python starts at 0 and `patch_schedule` is in order of operating system from OS 1 to OS 3, the index value 0 corresponds to the patch date for OS 1, the index value 1 corresponds to the patch date for OS 2, and so on.

Question 7 What happens when you run the code before modifying it? What is causing the logic errors? How can you fix them?

[When running the code as is, the output shows “Patch date: March 1st”. This may seem good with no errors at first, but when looking into the code, the system for “OS 2” should actually be assigned the date “April 1st”. To fix this logic error, the indices for each of the systems under the conditional statement “if”, needs to match the indices under “`patch_schedule`”. In the code above, we see that the indices under “if system ==”OS 1” and “elif system ==”OS 2“, do not show the correct indices for”`patch_schedule[]`“. To fix this, OS 1 should be corrected to”`patch_schedule[0]`“, and OS 2 should be corrected to”`patch_schedule[1]`“, to correct the logic error with the indices. When testing the code with each OS versions under”system“, the code outputs the correct dates and is running properly.]

1.10 Conclusion

What are your key takeaways from this lab?

[My key takeaways from this lab were learning how to correct syntax errors, logic errors, and exceptions within lines of codes, as well as being able to utilize the `print()` function to test codes and check if the errors were corrected. I learned that it is important to ensure that codes are written correctly with spelling and grammar, as well as making sure that the step-by-step procedures of codes logically flow with each other to produce the correct results.]